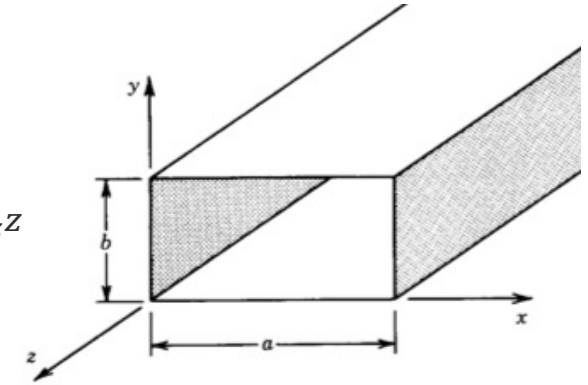
Metallic Waveguides

The following subjects will be covered

- Introduction.
- Wave Propagation in Metallic Waveguides.
- TE and TM Modes.
- Examples.

Wave Equation Solutions: TM Modes

Without going into complicated details, we can construct the required solutions by inspection.



1) TM Modes ($H_z = 0$)

$$E_z = (A\cos\beta_x x + B\sin\beta_x x)\cdot(C\cos\beta_y y + D\sin\beta_y y)\cdot e^{-j\beta_z z}$$

Apply boundary conditions, tangential fields=0 at $x = 0, a$ and $y = 0, b$.

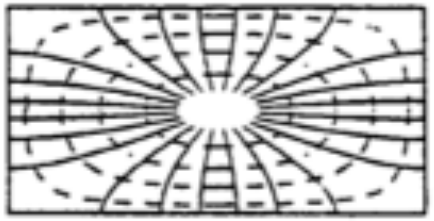
$$E_z = E_o \sin\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$m = 1, 2, \cdots \qquad n = 1, 2, \cdots$

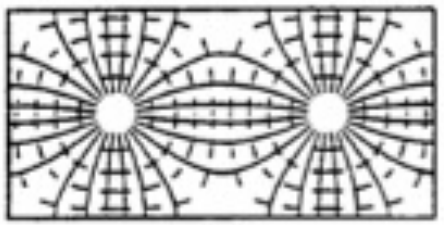
$$\beta_x = \frac{m\pi}{a} \qquad \beta_y = \frac{n\pi}{a}$$

Mode number is TM_{mn}

Modes TM_{00} and TM_{01}. and TM_{10} are not possible ($E_z = 0$)



The fundamental (dominant) mode



Wave Equation Solutions: TM Modes

To find other field components, use Maxwell's equations.

$$E_z = E_o \sin\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$E_x = \frac{-j\beta}{\beta_{xy}^2}\frac{m\pi}{a}E_o \cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

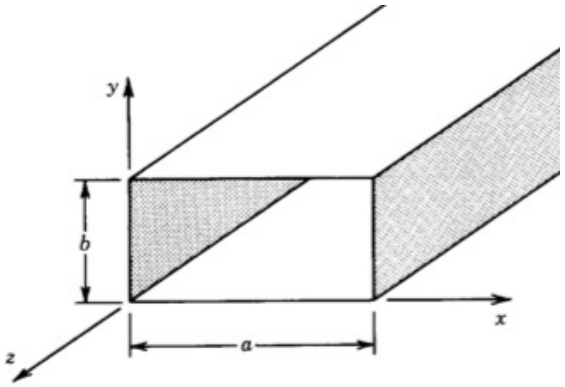
$$E_y = \frac{-j\beta}{\beta_{xy}^2}\frac{n\pi}{b}E_o \sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$H_x = \frac{j\omega\varepsilon}{\beta_{xy}^2}\frac{n\pi}{b}E_o \sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$H_y = \frac{-j\omega\varepsilon}{\beta_{xy}^2}\frac{m\pi}{a}E_o \cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$H_z = 0$$

$$\beta_{xy}^2 = \beta_x^2 + \beta_y^2$$



$$\beta = \sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

Should be real for mode propagation

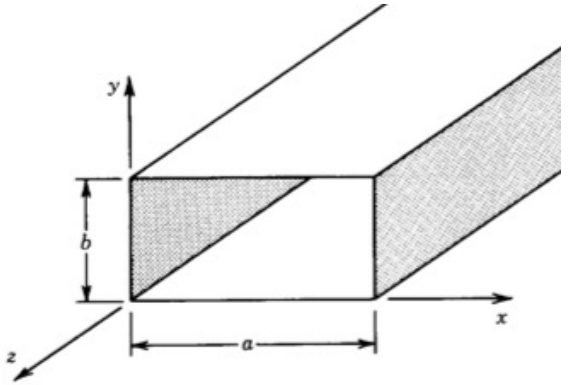
$$f_{mn} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

Cut-off (resonance) frequency

$$Z_{mn} = \eta\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}$$

Wave Equation Solutions: TE Modes

Without going into complicated details, we can construct the required solutions by inspection.



2) TE Modes ($E_z = 0$)

$$H_z = H_o \cos\left(\frac{m\pi}{a}x\right) \cdot \cos\left(\frac{n\pi}{b}y\right) \cdot e^{-j\beta z}$$

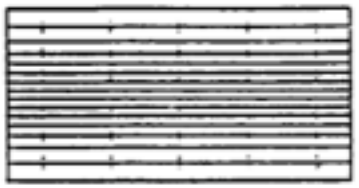
$m = 0, 1, 2, \cdots \qquad n = 0, 1, 2, \cdots$

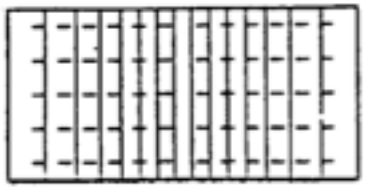
$$\beta_x = \frac{m\pi}{a} \qquad \beta_y = \frac{n\pi}{a}$$

Mode number is TE_{mn}

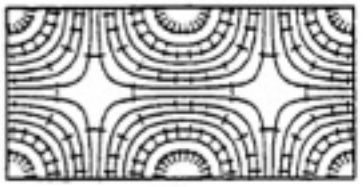
Mode TE_{00} is not possible

But modes TE_{01}. and TE_{10} are valid





The fundamental (dominant) mode (depends on a and b)



Wave Equation Solutions: TE Modes

To find other field components, use Maxwell's equations.

$$H_z = H_o \cos\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$H_x = \frac{j\beta}{\beta_{xy}^2}\frac{m\pi}{a}H_o \sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

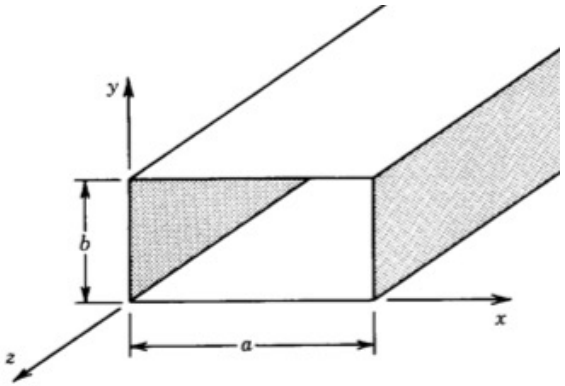
$$H_y = \frac{j\beta}{\beta_{xy}^2}\frac{n\pi}{b}H_o \cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$E_x = \frac{j\omega\mu}{\beta_{xy}^2}\frac{n\pi}{b}H_o \cos\left(\frac{m\pi}{a}x\right)\cdot\sin\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$E_y = \frac{-j\omega\mu}{\beta_{xy}^2}\frac{m\pi}{a}H_o \sin\left(\frac{m\pi}{a}x\right)\cdot\cos\left(\frac{n\pi}{b}y\right)\cdot e^{-j\beta z}$$

$$E_z = 0$$

$$\beta_{xy}^2 = \beta_x^2 + \beta_y^2$$



$$\beta = \sqrt{\omega^2\mu\varepsilon - \left(\frac{m\pi}{a}\right)^2 - \left(\frac{n\pi}{b}\right)^2}$$

Should be real for mode propagation

$$f_{mn} = \frac{1}{2\sqrt{\mu\varepsilon}}\sqrt{\left(\frac{m}{a}\right)^2 + \left(\frac{n}{b}\right)^2}$$

Cut-off (resonance) frequency

$$Z_{mn} = \frac{\eta}{\sqrt{1 - \left(\frac{f_{mn}}{f}\right)^2}}$$

Examples

1) Consider a RWG with dimensions 2.5 cm × 1 cm, filled with a lossless material of ε_r=4. Find the cutoff frequencies of all allowed modes up to 15 GHz. Which is the dominant mode?

$$f_{mn} = \frac{c}{4}\sqrt{\left(\frac{m}{0.025}\right)^2 + \left(\frac{n}{0.01}\right)^2}$$

11 TE modes
4 TM modes

Dominant mode:
TE_{10}

Single mode operation between 3-6 GHz

TE_{mn}	f_{mn} (GHz)
10	3
01	7.5
11	8.078
02	15
03	22.5
20	6
30	9
40	12
50	15
60	18

TM_{mn}	f_{mn} (GHz)
10	Not allowed
01	Not allowed
11	8.078
12	15.3
21	9.6
31	11.72
41	14.14

Examples

2) A TM wave is propagating in dielectric-filled RWG with dimensions $a = 2b$ =4 cm. The following field component is known.

$$H_y = 6\cos(25\pi\, x)\cdot\sin(100\pi\, y)\cdot\sin(1.5\pi\times10^{10}\, t - 109\pi\, z)\ \text{mA/m}$$

Find: 1) mode number, 2) ε_r of material, 3) phase velocity, 4) wave impedance and 5) E_x.

Compare expressions with standard form

$$\beta_x = \frac{m\pi}{a} = 25\pi\, x \rightarrow m = 1$$

$$\beta_y = \frac{n\pi}{b} = 100\pi\, x \rightarrow n = 2$$

$\Rightarrow TM_{12}$

From the β expression, solve for ε_r.

$$\beta = 109\pi = \sqrt{(1.5\pi\times10^{10})^2\mu\varepsilon - (25\pi)^2 - (100\pi)^2} \quad \rightarrow \quad \varepsilon_r = 9$$

The phase velocity is given by ω/β.

$$v = \frac{1.5\pi\times10^{10}}{109\pi} = 1.38\times10^{8} \quad \text{m/s}$$

Wave impedance for TM modes is

$$Z_{12} = \frac{377}{3}\sqrt{1 - \left(\frac{5.15\times10^{9}}{0.75\times10^{10}}\right)^2} = 91.3\ \Omega$$

To find E_x, note that $Z_{12} = \dfrac{|E_x|}{|H_y|}$

$$E_x = (0.006)(91.3)\cos(25\pi x)\cdot\sin(100\pi y)\cdot\sin(1.5\pi\times10^{10}\, t - 109\pi\, z)\ \text{V/m}$$

Q & A

Send me an email if you have any questions.